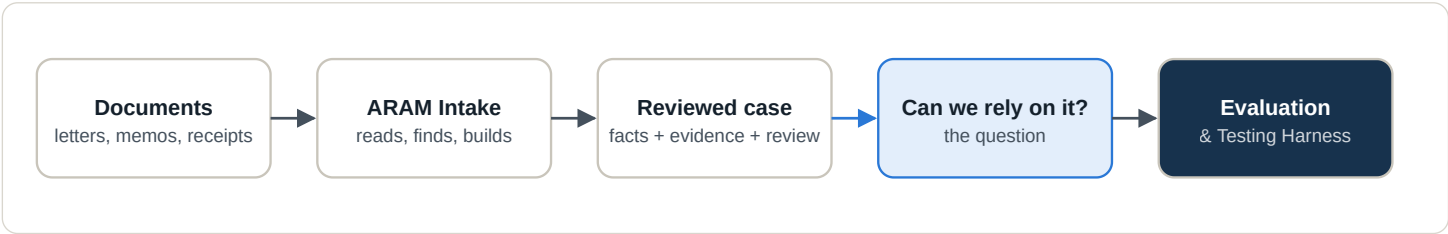


From Employment Documents to a Verified Case

How ARAM tests the reliability of Intake: independently, step by step, and in a way that makes every failure explainable.

How do we know Intake behaves correctly when the evidence is complete, incomplete, ambiguous or contradictory?

Baseline **golden-2026-09-25** Dataset **TAURI-DE-GOLDEN-40** Scored jobs **240** Fixtures **40**



Written for four questions

GERMAN HR

“Can I rely on the information presented for review?”

IN-HOUSE COUNSEL

“Can we show how a result was produced and tested?”

EXTERNAL COUNSEL

“Can I inspect the evidence behind a result, and understand a failure?”

EMPLOYMENT LAWYER

“Does the system preserve uncertainty instead of silently turning it into a fact?”

How to read this: Pages 1–12 explain the method with one synthetic Route D matter. Pages 13–15 report the real baseline **golden-2026-09-25**, weaknesses included. Each page has an *Engineer’s view* with the mechanism, identifiers and sources for the customer engineering team.

Before relying on a case, six questions matter

The evaluation is organised around the questions a professional would ask of any intake process. Each question is one evaluation layer, and each layer has its own evidence and metrics.

1

Read
Did ARAM recover the source information correctly?

L0

2

Understand
Did ARAM identify each document and route it to the right rule pack?

L1

3

Find
Did ARAM identify the expected facts, each with its evidence?

L2

.

Human verification
Did a person explicitly confirm, correct, reject or mark as unknown each proposed value?

H1

4

Build
Did facts, conflicts and reviewer decisions reach the right case fields?

L3

5

Check
Is the resulting case structurally complete and internally valid?

L4

6

Verify
Did the complete machine-plus-review workflow produce the expected final case?

L5

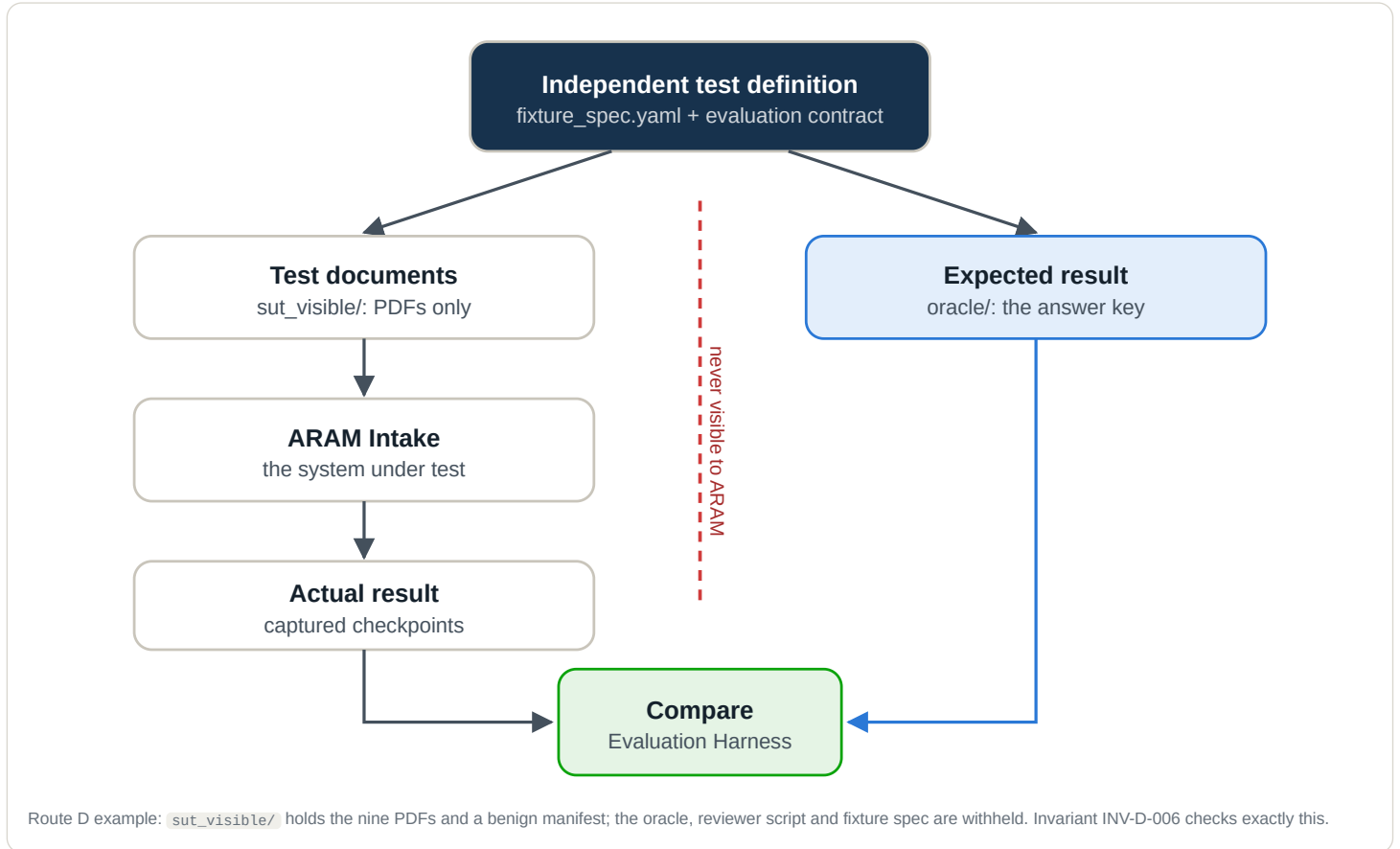
Engineer's view

Layer names follow the repository: L0 Document Recovery, L1 Classification, L2 Fact Extraction, **H1 Human Candidate Verification**, L3 Case Transformation, L4 Schema & Invariants, L5 End-to-End Product. H1 sits between L2 and L3: L2 candidates are `PROPOSED` and H1 turns them into `CONFIRMED` / `CORRECTED` / `REJECTED` / `UNKNOWN` . L3 consumes only H1-verified facts. L2 is scored *before* H1, and "human correction does not repair Layer 2 metrics".

Sources: `Case_Intake_Evals_and_Testing_harness/README.md` (evaluation architecture); `Evals_Layer_0 - 5/README_LAYER_2_Fact_Extraction.md` §3.4–3.5, §7.4.

ARAM takes the test. ARAM does not write its own answer key.

Every test case starts from one authoritative definition. From it, two things are produced separately: the documents ARAM is allowed to see, and the expected result, which ARAM never sees.



The expected result is derived independently from the frozen test-case definition and contract. The system under test never receives answer-key material.

Engineer's view

- **Oracle independence policy:** a fixture/oracle component “must not import, call, copy algorithmic outputs from, or reuse the exact production implementation of the behavior it scores.” Every Route D oracle file declares `sut_modules_imported: []`, `uses_tauri_predictions: false`, `uses_production_mapper_as_oracle: false`, `uses_production_extractor_as_oracle: false`. Only shared primitives are allowed (ISO-8601 parser, SHA-256).
- **SUT isolation:** never exposed to the SUT are `fixture_spec.yaml`, `oracle/`, golden approvals, expected labels/candidate IDs, forbidden-candidate lists, the validation report and hidden semantic tags. A leakage gate scans visible PDF text, PDF/XMP metadata, filenames, attachments, structured inputs and workspace paths.

Sources: TAURI_INTAKE_DATASET_AGENT_DESIGN.md §7.3, §34;

EVALS_DATASET_AND_GROUND_TRUTH_DESIGN.md §16, §22; CASE-RD-06824/oracle/*.yaml.

One system runs the test. Another marks it.

Running a test and scoring it are separate responsibilities, handled by two separate components, so that the record of what ARAM did cannot be shaded by how it will be scored.

Test Harness <i>"Run the test, and record it."</i> • Checks the test case is eligible and its hashes match • Gives ARAM only the permitted inputs • Runs the real Intake behaviour • Captures a checkpoint at every layer • Captures reviewer actions • Preserves all run artefacts under a run manifest It does not score.	Evaluation Harness <i>"Mark the test, and explain it."</i> • Validates the evaluation inputs • Compares expected vs actual • Calculates metrics with explicit numerator and denominator • Assigns a scoring status to every row • Classifies failures and finds their origin • Produces the reports and baseline comparisons It does not run the product.
---	---

The Test Harness records what ARAM actually did. The Evaluation Harness independently decides how that behaviour is scored.

Engineer's view

Run modes (Test Harness §3): `isolated` , `end_to_end` , `scripted_review` (L5-A), `interactive_review` (L5-B). Every run writes a manifest (`run_id` , `fixture_version` , `fixture_spec_hash` , `evaluation_contract_hash` , `intake_version` , `harness_version` , `injected_oracle_hashes` , ...). Scoring statuses and failure classification live only in the Evaluation Harness (§3, §12–14).

Sources: `TAURI_INTAKE_TEST_HARNESS_DESIGN.md` ; `TAURI_INTAKE_EVALUATION_HARNESS_DESIGN.md` .

A test case has to earn the right to judge ARAM

Before a test case can count, it has to pass seven gates. Before it can become part of the golden set, people have to approve every critical fact and every piece of provenance.

Seven acceptance gates

G0 Contract lock

The evaluation contract resolves and its hash matches; oracle independence is declared.

G1 Fixture-spec integrity

Schema passes; critical-fact authority is valid; IDs are unique; states are coherent.

G2 Source / build integrity

Required facts were injected exactly; no undeclared critical values slipped in.

G3 Rendered evidence integrity

Every required piece of evidence exists on the rendered page and matches.

G4 Mutation integrity

Any deliberate distortion (e.g. a scan artefact) is declared and its effect documented.

G5 Oracle integrity

The answer key is compiled independently; human-only decisions are not guessed.

G6 Isolation / replay / package

ARAM cannot see the answer key; the package is complete and replayable by hash.

Any mandatory gate failure removes the test case from product scoring. A broken test case cannot lower, or raise, ARAM's score.

Promotion ladder

GENERATED

Produced by the fixture generator. Not yet trusted.

VALIDATED_STRESS

Passed automated gates. Usable for stress testing.

GOLDEN

Every critical fact and its provenance reviewed and approved by people.

FROZEN_REGRESSION

Artefact hashes frozen. Admitted to regression testing.

"The agent cannot promote its own output." Golden approval requires an evaluation engineer and a domain reviewer, with 100% review of critical (Tier-1) facts and 100% review of their provenance. Any change to a reviewed artefact's hash invalidates the approval.

Engineer's view

- Gates G0–G6: Contract Lock · Fixture-Spec Integrity · Source/Build Integrity · Rendered Evidence Integrity · Mutation Integrity · Oracle Integrity · Isolation/Replay/Package Integrity.
- Full ladder: GENERATED → VALIDATED_STRESS → evaluation-engineering review → domain plausibility review → 100% Tier-1 fact review → 100% Tier-1 provenance review → expected case/review-state approval → GOLDEN → frozen artifact hashes → FROZEN_REGRESSION .
- Approval record approvals/golden_review.yaml , roles EVALUATION_ENGINEER and DOMAIN_REVIEWER . Compatibility states: ACTIVE_CURRENT , ACTIVE_LEGACY , DEPRECATED , QUARANTINED . Dataset splits: DISCOVERY , REGRESSION , HELD_OUT_VALIDATION , HUMAN_STUDY . "A fixture cannot silently change split."
- Tier-1 = a fact whose wrong value "can change route, gate, deadline, or verified-case correctness".

Sources: EVALS_DATASET_AND_GROUND_TRUTH_DESIGN.md §9, §18, §20–21; TAURI_INTAKE_DATASET_AGENT_DESIGN.md §39, §41–42; TAURI_INTAKE_EVAL_REGRESSION_CI_DESIGN.md §10–11.

One case. Seven points of verification.

Synthetic Route D matter `CASE-RD-06824` : an extraordinary termination for conduct of a pregnant employee, with the authority's approval obtained before the termination. The delivery evidence is deliberately defective.

The matter (synthetic)

Nine German documents: HR status record, investigation report, knowledge & deadline memo, decision memo, authority approval, termination letter, posting receipt, online tracking status, evidence register.

- DOC-001 personalakte_status
- DOC-002 untersuchungsbericht
- DOC-003 kenntnis_fristvermerk
- DOC-004 entscheidungsvermerk
- DOC-005 behoerdenzustimmung
- DOC-006 kuendigungsschreiben
- DOC-007 einlieferungsbeleg
- DOC-008 sendungsstatus
- DOC-009 evidenzregister

Key dates in the answer key

Employment start	2022-03-01
Triggering incident	2026-04-02
Complete knowledge of decision-maker	2026-04-07 11:20
Authority approval (pregnancy)	2026-04-17
Termination issued	2026-04-18 16:10
Two-week deadline (§ 626 BGB)	2026-04-21
Tracking says "ZUGESTELLT"	2026-04-20 (not proof of receipt)

L0 Did ARAM recover the source text and values faithfully?

WHAT IS INSPECTED

- Critical text spans and values
- Page recovery and association
- Correct recovery path
- Stopping safely when a page cannot be read

IN THIS CASE

The knowledge timestamp `2026-04-07T11:20:00+02:00` must be recovered exactly from DOC-003, page 1.

BASELINE RESULT · CASE-GD-06824

- Critical Text Recovery Rate: 100.0%
- Critical Value Accuracy: 100.0%
- Page Recovery & Association Accuracy: 100.0%
- Recovery Path Accuracy: 100.0%

Engineer's view

Oracle: `oracle/l0_recovery.yaml`. Metric #5 Unsafe Continuation Rate (`unsafe_continuations / fail_closed_opportunities`) had no eligible units in this baseline: none of the 40 fixtures contained a fail-closed opportunity.

L1 Did ARAM recognise each document and send it to the right rule pack?

WHAT IS INSPECTED

- Document type per document
- Confidently wrong labels
- Rule-pack routing
- Abstaining or flagging when a document is genuinely ambiguous

IN THIS CASE

All nine documents are expected to route to `extraordinary_termination_rule_pack`. DOC-008 must be recognised as `POSTAL_TRACKING_STATUS`, a tracking screen and not a delivery record.

BASELINE RESULT · CASE-GD-06824

- Classification Accuracy: 41.7%
- Wrong-Confident Classification Rate :: 37.5%
- Wrong Rule-Pack Routing Rate :: 50.0%
- Correct Abstention / Ambiguity Rate: 50.0%

Engineer's view

Oracle: `oracle/l1_classification.yaml` (`expected_type`, `expected_routing`, `ambiguity_required`). Isolated L1 runs receive the known-good L0 output, so an L1 score is not contaminated by L0 errors.

L2 Did ARAM find the expected facts, with evidence, and hold back where it must?

WHAT IS INSPECTED

- Expected facts found (recall) and nothing spurious (precision)
- Correct values
- Evidence trace for every candidate
- Safe abstention where a rule forbids a conclusion

IN THIS CASE

The tracking date 2026-04-20 is an expected candidate (CAND-071). The answer key also lists forbidden candidates:
termination_received_at = 2026-04-20 and delivery_proof = true .

BASELINE RESULT · CASE-GD-06824

Fact Recall: 42.9%

Fact Precision: 54.5%

Value Accuracy: 100.0%

Safe Rule Abstention Rate: 100.0%

Candidate Evidence Traceability Rate: 75.0%

Engineer's view

Oracle: oracle/l2_candidates.yaml with forbidden_candidates . Fact Recall = matched_expected_candidates / expected_extractable_candidates ; Fact Precision = matched_actual_candidates / scored_actual_candidates . L2 is stored and scored before H1.

H1 Does every proposed value receive an explicit human decision?

WHAT IS INSPECTED

- Per-value disposition: confirm, correct, reject, mark unknown
- Source span inspected
- Conflicting candidates compared

IN THIS CASE

Scripted review for this case: REV-071 REJECT delivery_date ("Tracking status cannot establish receipt.") and REV-077 MARK_UNKNOWN termination_received_at .

BASELINE RESULT · CASE-GD-06824

H1 has no standalone metric. Its effect is measured at L3 (confirmed facts preserved) and L5 (errors that escape review).

Engineer's view

"Every proposed document-derived value requires explicit per-value H1 disposition before it may become a confirmed intake fact." L3 diagnostics include "H1 Correction Override Rate" and "H1 Authority Violation Rate" (target 0%). Source: layer READMEs L2 §7.4, L3 §8, §12.

L3 Did confirmed facts, conflicts and review states reach the right case fields unchanged?

WHAT IS INSPECTED

- Field mapping
- Confirmed facts preserved
- Derived fields
- Conflicts stay visible; no silent overwrite; lineage kept

IN THIS CASE

11 expected fields. termination_received_at must stay REVIEW_REQUIRED with the reason "Tracking assertion is not accepted as receipt proof". The lineage of the rejected tracking candidate must be retained.

BASELINE RESULT · CASE-GD-06824

Field Mapping Accuracy: 87.5%

Confirmed-Fact Preservation Rate: 85.7%

Transformation Completeness: 87.5%

Conflict & Provenance Safety: 100.0%

Engineer's view

Oracle: oracle/l3_transformation.yaml . Isolated L3 input = oracle/l2_candidates.yaml + reviewer/scripted_actions.yaml . Conflict & Provenance Safety = passed_conflict_and_provenance_assertions / applicable_safety_assertions .

L4 Is the case structurally valid and internally consistent?

WHAT IS INSPECTED

- Schema validity
- Required fields present
- Invariants (rules that must always hold)
- Evidence states valid

IN THIS CASE

- INV-D-001 2026-04-07 + 14 days = 2026-04-21
- INV-D-003 approval 2026-04-17 predates issue 2026-04-18
- INV-D-004 tracking date is not promoted to confirmed receipt

BASELINE RESULT · CASE-GD-06824

- Schema Pass Rate: 100.0%
- Required-Field Completeness: 100.0%
- Evidence-State Validity: 100.0%

Engineer's view

Oracle: oracle/l4_case_and_assertions.yaml (6 invariants INV-D-001...006, evidence states ED-11 and EU-WF DEFECTIVE). Note: metric #23 Invariant Pass Rate had no eligible units across all 40 golden fixtures in this baseline. Engineering should confirm whether invariant assertions are wired into the golden set.

L5 Did the full machine-plus-review workflow produce exactly the expected verified case?

WHAT IS INSPECTED

- Strict pass of the whole case
- Critical (Tier-1) field accuracy
- Critical errors that escape review

IN THIS CASE

A strict pass requires all Tier-1 values and states to be correct, termination_received_at to remain REVIEW_REQUIRED , delivery_proof to remain false , and no critical conflict to be silently discarded.

BASELINE RESULT · CASE-GD-06824

- Verified Case Pass Rate: 0.0%
- Critical Field Accuracy: 50.0%
- Critical Error Escape Rate : 83.3%
- Scripted Review Action Burden: 9.1%

Engineer's view

Oracle: oracle/l5_case_and_review.yaml , review_population: SCRIPTED_L5_A . Human Correction Burden: NOT_SCORED ("L5-A scripted regression is not real-human telemetry"). Scripted review action burden is DIAGNOSTIC_ONLY .

About the pairing: Documents and expected values come from dataset fixture CASE-RD-06824 (status VALIDATED_STRESS, not GOLDEN). The results shown are for CASE-GD-06824 , the Route D fixture with the same case number in the golden baseline. The repository does not document how the two versions relate, so read the pairing as illustrative.

A fact must not become detached from its evidence.

For every confirmed value, the answer key records exactly where it comes from, down to the page and the region on the page. The evaluation checks that ARAM keeps this chain intact.

CASE FIELD
knowledge_date
✓ CONFIRMED 2026-04-07T11:20:00+02:00

↓

EXTRACTED CANDIDATE
CAND-026 · ASSERT-026
Proposed at L2, confirmed at H1

↓

SOURCE REFERENCE
DOC-003 · page 1 · ANCHOR-DOC-003-001
bbox [204.09, 665.82, 313.34, 677.82] · Kenntniszeitpunkt

↓

PINNED SOURCE ARTEFACT
sha256:e31d98e885dd17acc25e9022fe68ed22929bcd0e21d718d0f824fe9cf2ec0386
SHA-256 of the rendered document

Kenntnis- und Fristvermerk

2026-04-07 | DOC-003 | CASE-RD-06824

Der Fristvermerk trennt Ereignisdatum, vollständige positive Kenntnis des kündigungsberechtigten Entscheidungsträgers und berechnete Zwei-Wochen-Frist.

Kenntniszeitpunkt	2026-04-07T11:20:00+02:00
Kündigungsberechtigte Stelle	PERSON-HR-01 (Leitung Personal)
Kenntniskette	Compliance review -> HR case conference -> authorized decision-maker received the complete evidence packet.
Zwei-Wochen-Frist endet	2026-04-21
Kündigung ausgestellt am	2026-04-18T16:10:00+02:00
Frist eingehalten	Ja / true

DOC-003 "Kenntnis- und Fristvermerk", page 1. The highlighted region is drawn from the answer key's bounding box, not placed by hand.

HR
Where did this value come from?

In-house counsel
Can the organisation explain the evidence path?

External counsel / lawyer
Can the asserted fact be traced back to its source?

Engineer's view
Provenance record oracle/provenance.yaml: anchor_id, assertion_id, candidate_id, page, bbox (PDF_POINTS_BOTTOM_LEFT), source_text, artifact_hash . Measured by L2 #15 Candidate Evidence Traceability Rate (candidates_with_valid_trace / candidates_requiring_provenance) and L3 #20 Conflict & Provenance Safety.

The evaluation tests uncertainty, not just easy answers.

In this matter, the online tracking screen says “ZUGESTELLT” on 2026-04-20. There is no delivery record and no identified delivery person. A system that simply copied the date would manufacture a receipt date that the evidence does not support.

Online-Sendungsstatus

2026-04-20 | DOC-008 | CASE:RD-06824

Synthetische Statusanzeige. Der Status behauptet eine Zustellung, enthält aber keinen Auslieferungsbeleg und keine identifizierte Zustellperson.

Sendungsnummer

RR041826510DE

Status

ZUGESTELLT

Statusdatum

2026-04-20

Auslieferungsbeleg vorhanden

Nein / false

DOC-008 “Online-Sendungsstatus”. Red: the status date, a candidate that must *not* become the receipt date. Amber: “Auslieferungsbeleg vorhanden: Nein / false”.

What the answer key expects

Receipt stays open, for a human to decide, instead of becoming a fact.

FIELD	EXPECTED STATE (ANSWER KEY)	
delivery_method	✓ CONFIRMED	REGISTERED_MAIL
delivery_proof	✓ CONFIRMED	false
termination_received_at	🕒 REVIEW_REQUIRED	null

✗ Fails Silent choice

- termination_received_at = 2026-04-20 taken from the tracking screen
- delivery_proof = true
- The conflict disappears from the case

✓ Passes Preserve the uncertainty

- Tracking date kept as a candidate, with its source
- Receipt remains REVIEW_REQUIRED / unknown
- Reviewer rejects the date and marks receipt unknown
- Rejected candidate lineage retained

Important wording. This demonstrates *tested system behaviour* around evidence, uncertainty, provenance and review state. It does not show that any legal conclusion about receipt, deadlines or validity of a termination is correct.

Engineer's view

- forbidden_candidates in oracle/l2_candidates.yaml: termination_received_at = 2026-04-20 (“Online tracking status is not authoritative proof of actual receipt in this fixture.”) and delivery_proof = true .
- Evidence profile: ED-11 and EU-WF = PROVENANCE_INVALID / DEFECTIVE .
- Invariants INV-D-004 and INV-D-005 .
- A silently discarded critical conflict is a zero-tolerance CI blocker.

Review is part of the workflow, and scripted review is kept apart from real human review.

ARAM proposes; a person decides. The evaluation measures this in three distinct places and never pools them.

- Confirm value is right
- Correct value replaced
- Reject candidate discarded
- Resolve conflict decided
- Mark unknown evidence insufficient

H1

Human candidate verification

Every proposed value needs an explicit decision before it becomes a confirmed fact.

Scored indirectly at L3 and L5. Never used to repair L2 scores.

L5-A

Scripted regression review

A fixed, reproducible reviewer script drives the full product flow. It shows the workflow behaves the same way run after run.

CASE-RD-06824 script: 77 actions

✓ CONFIRM × 75

✗ REJECT × 1

👁 MARK_UNKNOWN × 1

Scripted Review Action Burden (diagnostic)

17.3%_{69/398}

L5-B

Real human reviewer study

Real reviewers under a declared protocol. This is the only population that can show how much correction work people actually do.

Human Correction Burden ↓

not computed: no eligible units

No L5-B study is part of this baseline.

Never merged. Human Correction Burden belongs only to the real-human L5-B study. In this baseline it is “—” (no eligible units), not 0%. The scripted run reports a separately named diagnostic instead.

Engineer's view

Run modes: `scripted_review` (L5-A, deterministic reviewer scripts) vs `interactive_review` (L5-B, real reviewer telemetry). $\text{Human Correction Burden} = (\text{corrected_machine_suggestions} + \text{manual_field_entries} + \text{manually_resolved_conflicts}) / \text{reviewed_fields}$, L5-B only.

`scripted_review_action_burden` "must not use the canonical Human Correction Burden metric id".

Critical Error Escape Rate is reported separately for L5-A and L5-B.

A wrong final case is not enough information.

End-to-end testing shows that a case failed. Isolated testing gives each layer known-good input from the answer key, so each capability is measured on its own.

LAYER UNDER TEST	RECEIVES KNOWN-GOOD INPUT	WHAT THIS ISOLATES
L1	Recovered text from the answer key oracle/l0_recovery.yaml	Classification quality, independent of reading errors
L2	Recovered text + correct classification oracle/l0_recovery.yaml + oracle/l1_classification.yaml	Extraction quality, independent of reading and routing errors
L3	Correct candidates + scripted reviewer decisions oracle/l2_candidates.yaml + reviewer/scripted_actions.yaml	Transformation quality, independent of extraction and review errors
L4	Correct transformed case state oracle/l3_transformation.yaml	Structural validity, independent of upstream errors
L5	Nothing injected: the complete workflow full workflow	What a user actually experiences

End-to-end testing tells us that a case failed. Isolated testing helps identify which capability introduced the failure.

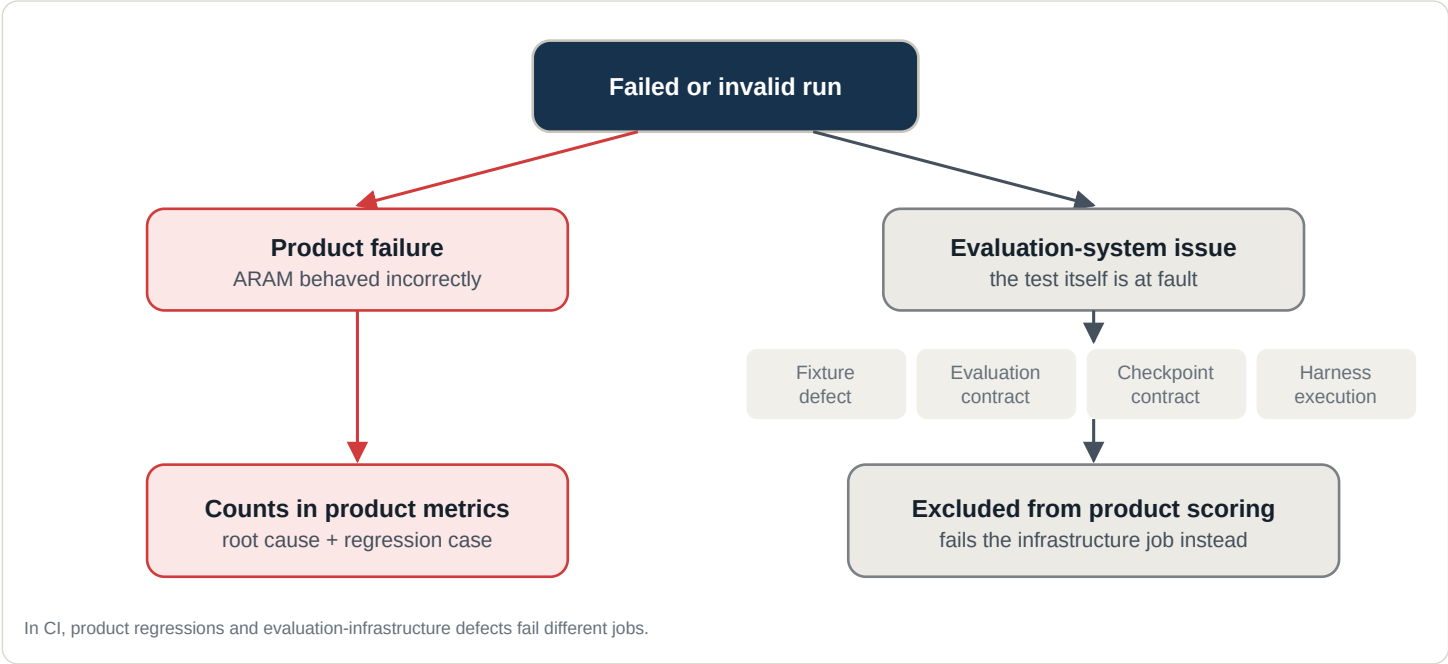
Root cause is assigned to the **earliest causally sufficient divergence**, found by walking the recorded lineage, not by assuming the earliest layer is to blame. Contributing causes are kept separately. For example: if L0 misreads a date and a reviewer then accepts it, the primary cause is L0 and `HUMAN_REVIEW_ESCAPE` is recorded as an additional cause.

Engineer's view

"These are compiled interfaces, not independently authored truth files. Every injected artifact is hash-pinned to the fixture manifest and evaluation-contract version" (Test Harness §5). Injected hashes appear in the run manifest as `injected_oracle_hashes`. "Report isolated layer quality separately from end-to-end observed quality" (L0–L5 design §6). Causal rule: "Do not infer causality from layer ordering alone" (§11).

A broken test must not be presented as a product failure.

Every scored row carries a status. Only rows marked `SCORED` enter a product metric. A defective test case, a contract gap or a harness crash is diagnosed separately and never becomes a zero in ARAM's score.



A real example: Route C

The Route C fixture's evaluation contract says certain social-selection evidence (EC-2 to EC-6) is always required. The underlying scenario, a complete closure, says social selection was unnecessary. Rather than invent a selection pool to satisfy the contract, the fixture marks the affected assertions as not scored until the contract owner resolves the gap (CG-001, with a second gap, CG-002).

```
route_c_2_AZR_276_16_dataset/oracle/l4_case_and_assertions.yaml →
NOT_SCORED_CONTRACT_ERROR_FOR_AFFECTED_ASSERTIONS
```

In this baseline

All 240 jobs in `golden-2026-09-25` were `SCORED`: nothing was excluded. Where a metric had nothing to measure, it is shown as `—`, not as 0%.

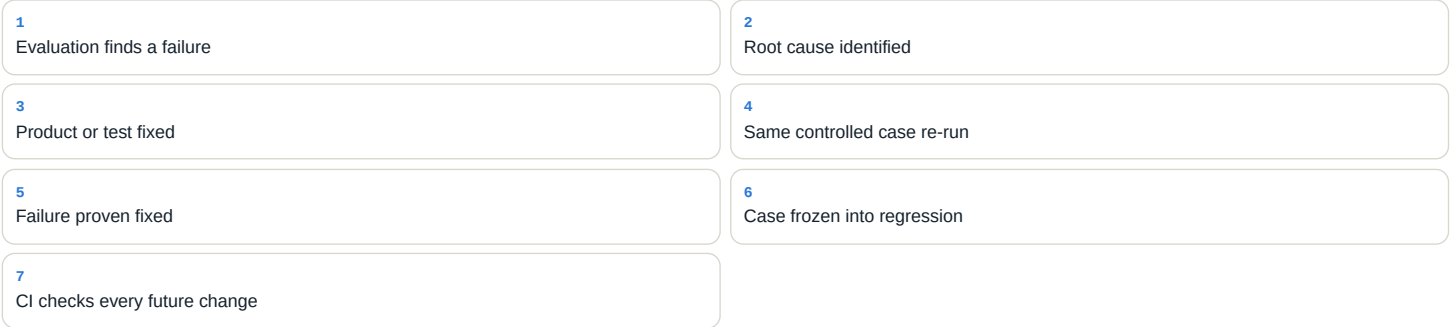
Engineer's view

Statuses: `SCORED`, `NOT_APPLICABLE`, `NOT_SCORED_EXECUTION_ERROR`, `NOT_SCORED_CONTRACT_ERROR`, `NOT_SCORED_FIXTURE_DEFECT`. "A fixture defect does not lower product metrics, but CI should fail the evaluation infrastructure job when a required regression fixture becomes unscorable." Quarantine flow: mark `NOT_SCORED_FIXTURE_DEFECT` → exclude from denominators → move to `QUARANTINED`. "Do not assign product root-cause layers to unscorable fixtures."

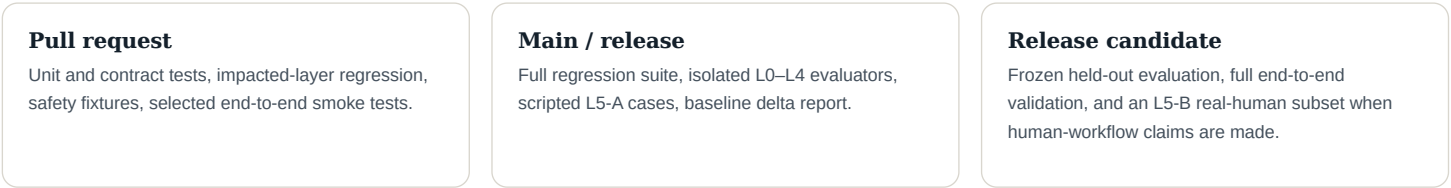
Sources: `TAURI_INTAKE_EVALUATION_HARNESS_DESIGN.md` §3, §13; `TAURI_INTAKE_EVAL_REGRESSION_CI_DESIGN.md` §7, §12; `Route C README.md`, `fixture_spec.yaml` (`GENERATED_CONTRACT_REVIEW_REQUIRED`), `approvals/golden_review.yaml` (`PENDING_HUMAN_REVIEW`).

A discovered failure becomes a permanent regression check.

The evaluation produces replayable evidence, not one-off demonstrations. Every run is pinned to exact versions and hashes, so a result can be reproduced and a fix can be proven.



Three CI tiers



Zero-tolerance product blockers (any new regression)

- ✗ Tier-1 correctness
- ✗ Unsafe continuation
- ✗ Silent overwrite
- ✗ Critical invariant failure
- ✗ Wrong rule pack reaching extraction
- ✗ Silently discarded critical conflict
- ✗ Scripted L5-A safety escape
- ✗ Schema regression on frozen fixtures

Every run is pinned

<code>run_id</code> this run	<code>fixture_version</code> test-case version	<code>fixture_spec_hash</code> test-case definition	<code>evaluation_contract_hash</code> scoring contract	<code>intake_version</code> ARAM build	<code>harness_version</code> harness build	<code>injected_oracle_hashes</code> known-good inputs
---------------------------------	---	--	---	---	---	--

Engineer's view
Replay modes: `BYTE_REPLAYABLE` , `SEMANTICALLY_REPLAYABLE` , `FROZEN_ONLY` . "A replay creates a new run id and links `replay_of` ." Baseline comparison reports `new_failures` , `fixed_failures` , `persistent_failures` , `fixture_defects` , `contract_errors` , `metric_deltas` . "Do not compare incompatible semantics as if they were the same baseline."

Sources: `TAURI_INTAKE_EVAL_REGRESSION_CI_DESIGN.md` §3–4, §6, §9;
`TAURI_INTAKE_TEST_HARNESS_DESIGN.md` §8, §10.

Baseline golden-2026-09-25: what was measured

These are the measured results of the first golden baseline. Every value is shown with its numerator and denominator. Metrics that had nothing to measure are marked “—” and never shown as 0%.

Source Evals_Illustrative_docs/metrics_golden-2026-09-25.md — values, n/d and direction copied by script, not re-typed.

Evaluation context

BASELINE golden-2026-09-25	CREATED 2026-09-25T04:54:38Z
DATASET TAURI-DE-GOLDEN-40	SUITE main
JOBS SCORED 240 / 240 SCORED	EXCLUDED / NOT SCORED 0
FIXTURES 40 · A 5 · B 3 · C 7 · D 25	INTAKE BUILD · CONTRACT HASH · HARNESS VERSION not stated in the source report

How to read the numbers. Percent = pooled numerator ÷ pooled denominator over the jobs whose metric was computed (unit-weighted). ↓ marks a rate where lower is better. The report is descriptive; it does not by itself set pass/fail gates.

Results by customer question

■ higher is better ■ ↓ lower is better ▨ — not computed (not 0%)

L0 Read: Document Recovery Were source documents recovered faithfully?

Critical Text Recovery Rate #1 · l0.critical_text_recovery_rate		100.0%	404/404
Critical Value Accuracy #2 · l0.critical_value_accuracy		100.0%	404/404
Page Recovery & Association Accuracy #3 · l0.page_recovery_association_accuracy		100.0%	784/784
Recovery Path Accuracy #4 · l0.recovery_path_accuracy		100.0%	380/380
Unsafe Continuation Rate ↓ lower is better #5 · l0.unsafe_continuation_rate		—	not computed: no eligible units

L1 Understand: Classification & Routing Were documents recognised and routed correctly?

Classification Accuracy #6 · l1.classification_accuracy		59.2%	197/333
Wrong-Confident Classification Rate ↓ lower is better #7 · l1.wrong_confident_classification_rate		21.2%	53/250
Wrong Rule-Pack Routing Rate ↓ lower is better #8 · l1.wrong_rule_pack_routing_rate		39.5%	150/380
Correct Abstention / Ambiguity Rate #9 · l1.correct_abstention_ambiguity_rate		46.8%	22/47
Per-Type Precision & Recall #10 · l1.per_type_precision_recall		—	computed per type: no single ratio

L2 Find: Fact Extraction *Were the expected facts found, with evidence?*

Fact Recall #11 · l2.fact_recall		42.9%	188/438
Fact Precision #12 · l2.fact_precision		55.1%	188/341
Value Accuracy #13 · l2.value_accuracy		100.0%	212/212
Safe Rule Abstention Rate #14 · l2.safe_rule_abstention_rate		99.7%	617/619
Candidate Evidence Traceability Rate #15 · l2.candidate_evidence_traceability_rate		88.7%	188/212

L3 Build: Case Transformation *Were facts, conflicts and review states carried into the case safely?*

Field Mapping Accuracy #16 · l3.field_mapping_accuracy		94.8%	273/288
Confirmed-Fact Preservation Rate #17 · l3.confirmed_fact_preservation_rate		84.0%	368/438
Derived-Field Accuracy #18 · l3.derived_field_accuracy		—	not computed: no eligible units
Transformation Completeness #19 · l3.transformation_completeness		94.8%	273/288
Conflict & Provenance Safety #20 · l3.conflict_provenance_safety		100.0%	204/204

L4 Check: Schema & Invariants *Is the resulting case structurally valid and consistent?*

Schema Pass Rate #21 · l4.schema_pass_rate		82.5%	33/40
Required-Field Completeness #22 · l4.required_field_completeness		100.0%	486/486
Invariant Pass Rate #23 · l4.invariant_pass_rate		—	not computed: no eligible units
Evidence-State Validity #24 · l4.evidence_state_validity		100.0%	1,190/1,190

L5 Verify: End-to-End Product *Did the complete workflow produce the expected verified case?*

Verified Case Pass Rate #25 · l5.verified_case_pass_rate		2.5%	1/40
Critical Field Accuracy #26 · l5.critical_field_accuracy		63.9%	235/368
Critical Error Escape Rate #27 · l5.critical_error_escape_rate		63.0%	133/211
Human Correction Burden #28 · l5.human_correction_burden		—	not computed: no eligible units
Scripted Review Action Burden #D1 · l5a.scripted_review_action_burden		17.3%	69/398

There is deliberately no overall score. These metrics measure different stages, have different denominators (from 40 cases to 1,190 assertions), point in different directions, and some were not computed. Averaging them into a single number would mislead.

By route

Route D contributes 25 of the 40 fixtures, so pooled values mostly reflect Route D. Route B has only 3 fixtures; read its percentages with care.

Metric	Route A 5 fixtures	Route B 3 fixtures	Route C 7 fixtures	Route D 25 fixtures
Critical Text Recovery Rate #1	100.0% 40/40	100.0% 31/31	100.0% 75/75	100.0% 258/258
Critical Value Accuracy #2	100.0% 40/40	100.0% 31/31	100.0% 75/75	100.0% 258/258
Page Recovery & Association Accuracy #3	100.0% 77/77	100.0% 62/62	100.0% 174/174	100.0% 471/471
Recovery Path Accuracy #4	100.0% 37/37	100.0% 31/31	100.0% 99/99	100.0% 213/213
Classification Accuracy #6	71.4% 20/28	68.0% 17/25	71.8% 56/78	51.5% 104/202
Wrong-Confident Classification Rate ↓ #7	20.0% 5/25	37.0% 10/27	23.3% 17/73	16.8% 21/125
Wrong Rule-Pack Routing Rate ↓ #8	35.1% 13/37	32.3% 10/31	27.3% 27/99	46.9% 100/213
Correct Abstention / Ambiguity Rate #9	44.4% 4/9	16.7% 1/6	47.6% 10/21	63.6% 7/11
Fact Recall #11	56.8% 25/44	47.1% 16/34	49.4% 40/81	38.4% 107/279
Fact Precision #12	62.5% 25/40	64.0% 16/25	58.0% 40/69	51.7% 107/207
Value Accuracy #13	100.0% 28/28	100.0% 18/18	100.0% 42/42	100.0% 124/124
Safe Rule Abstention Rate #14	96.7% 59/61	100.0% 52/52	100.0% 149/149	100.0% 357/357
Candidate Evidence Traceability Rate #15	89.3% 25/28	88.9% 16/18	95.2% 40/42	86.3% 107/124
Field Mapping Accuracy #16	100.0% 34/34	100.0% 21/21	100.0% 54/54	91.6% 164/179
Confirmed-Fact Preservation Rate #17	84.1% 37/44	85.3% 29/34	82.7% 67/81	84.2% 235/279
Transformation Completeness #19	100.0% 34/34	100.0% 21/21	100.0% 54/54	91.6% 164/179
Conflict & Provenance Safety #20	100.0% 25/25	100.0% 17/17	100.0% 44/44	100.0% 118/118
Schema Pass Rate #21	100.0% 5/5	100.0% 3/3	85.7% 6/7	76.0% 19/25
Required-Field Completeness #22	100.0% 59/59	100.0% 38/38	100.0% 79/79	100.0% 310/310
Evidence-State Validity #24	100.0% 149/149	100.0% 77/77	100.0% 224/224	100.0% 740/740
Verified Case Pass Rate #25	20.0% 1/5	0.0% 0/3	0.0% 0/7	0.0% 0/25
Critical Field Accuracy #26	86.4% 38/44	66.7% 18/27	66.2% 45/68	58.5% 134/229
Critical Error Escape Rate ↓ #27	28.6% 6/21	64.3% 9/14	59.0% 23/39	69.3% 95/137
Scripted Review Action Burden #D1	19.6% 9/46	13.8% 4/29	13.3% 10/75	18.5% 46/248

What the baseline shows, candidly

The harness is working as intended: it shows precisely where ARAM Intake is reliable today and where it is not yet ready.

✓ Strong evidence

Critical Text Recovery Rate: **100.0%** (404/404)
Critical text recovered in every case.

Value Accuracy: **100.0%** (212/212)
When a fact is found, its value is right.

Conflict & Provenance Safety: **100.0%** (204/204)
Conflicts stayed visible; no silent overwrite.

Safe Rule Abstention Rate: **99.7%** (617/619)
Held back where rules required it (2 misses in 619).

Required-Field Completeness: **100.0%** (486/486)

Evidence-State Validity: **100.0%** (1,190/1,190)

✗ Needs improvement

Verified Case Pass Rate: **2.5%** (1/40)
Only 1 of 40 cases passed strictly end-to-end.

Critical Error Escape Rate ↓: **63.0%** (133/211)
Of critical machine errors present at review entry, 63% were still wrong or unresolved at exit.

Classification Accuracy: **59.2%** (197/333)

Wrong Rule-Pack Routing Rate ↓: **39.5%** (150/380)
Misrouted documents enter the wrong rule pack.

Fact Recall: **42.9%** (188/438)
Fewer than half of the expected facts were found.

Schema Pass Rate: **82.5%** (33/40)
7 of 40 cases failed schema validation.

! Interpret carefully

Route D is 25 of 40 fixtures and dominates pooled values; Route B has 3.

Five metrics were not computed (no eligible units, or no single ratio). That is not the same as 0%.

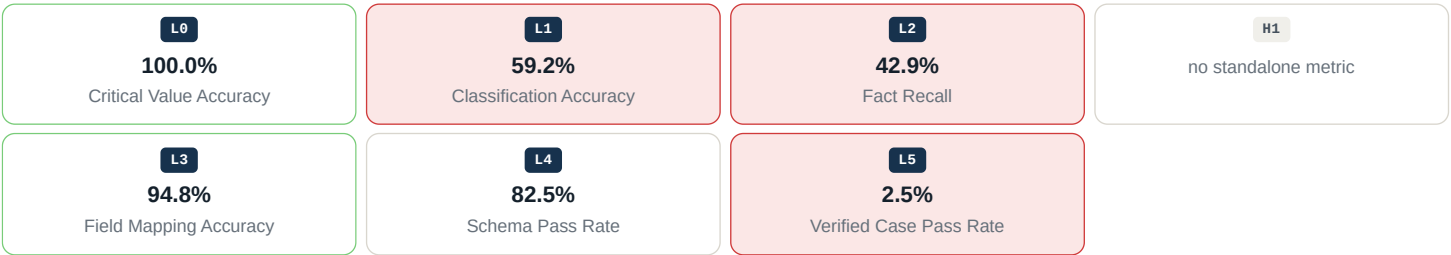
No real-human (L5-B) data: reviewer effectiveness and correction burden are unmeasured.

The end-to-end review in this baseline is scripted (L5-A diagnostic computed for all 40 cases). Real reviewers may catch more, or fewer, errors.

Intake build, contract hash and harness version are not stated in the report.

Where the weakness begins

Reading the layers in order shows a clear pattern: reading is clean, understanding and finding are the earliest weak layers, and the end-to-end result fails downstream.



This is the pattern visible in the report's layer metrics. The failure-by-failure root-cause records (earliest causally sufficient divergence per failure) are not part of this report, so no failure counts per family are shown here.

Exclusions

0 of 240 jobs were excluded. No fixture defects, contract errors or execution errors affected this baseline.

Regression status



This is the first golden baseline, so there is nothing to compare against yet. New, fixed and persistent failures are reported from the next compatible baseline onward.

Engineer's view

Weak-layer figures (n/d): Classification Accuracy 197/333; Wrong-Confident Classification Rate ↓ 53/250; Wrong Rule-Pack Routing Rate ↓ 150/380; Fact Recall 188/438; Fact Precision 188/341. L3 metrics are computed on what reached L3 (Field Mapping Accuracy 273/288), which is why they can be high while L5 is low. Not computed: #5, #10, #18, #23, #28. Worth an engineering follow-up: Invariant Pass Rate had no eligible units although Route D fixtures declare invariants (INV-D-001...006).

What this evidence does and does not establish

An assurance report is only as useful as its limits are clear.

✓ Does establish

- The answer key is independent of the system under test, and ARAM cannot see it.
- The real Intake behaviour was exercised and its outputs captured at every layer.
- Evidence provenance, conflicts, uncertainty and review states are part of the test.
- Test-case defects cannot silently turn into product failures.
- Runs are versioned and designed to be reproducible.
- Baseline `golden-2026-09-25` measured 40 synthetic German termination matters across four routes, with the results on pages 13–14.

✗ Does not establish

- That ARAM's output is legally correct, compliant or equivalent to a lawyer's work.
- How real reviewers perform: no L5-B study is included.
- Readiness for reliance without human review. At 2.5% strict end-to-end pass and 63.0% critical error escape, this baseline does **not** support that.
- Performance on real, non-synthetic matters or on routes not in the dataset.
- Any trend over time: this is the first baseline.

The purpose of the Evaluation & Testing Harness is not merely to produce a score. It creates reproducible evidence of what was tested, what ARAM actually did, how that compared with an independent expected result, where failures began, and whether a fix stays protected.

Metric catalogue with baseline values

All 28 catalogue metrics and the L5-A diagnostic, with the numerator/denominator definition from the metrics reference and the value from baseline `golden-2026-09-25`.

#	LAYER	METRIC	NUMERATOR / DENOMINATOR	VALUE	
1	L0	Critical Text Recovery Rate l0.critical_text_recovery_rate	correctly_recovered_critical_spans / expected_critical_spans	100.0%	
2	L0	Critical Value Accuracy l0.critical_value_accuracy	correct_critical_values / expected_critical_values	100.0%	
3	L0	Page Recovery & Association Accuracy l0.page_recovery_association_accuracy	correct_page_associations / expected_page_associations	100.0%	
4	L0	Recovery Path Accuracy l0.recovery_path_accuracy	correct_recovery_paths / recovery_path_cases	100.0%	
5	L0	Unsafe Continuation Rate ↓ l0.unsafe_continuation_rate	unsafe_continuations / fail_closed_opportunities	—	not computed:
6	L1	Classification Accuracy l1.classification_accuracy	correct_predictions / eligible_classifiable_documents	59.2%	
7	L1	Wrong-Confident Classification Rate ↓ l1.wrong_confident_classification_rate	wrong_high_confidence_predictions / high_confidence_predictions	21.2%	
8	L1	Wrong Rule-Pack Routing Rate ↓ l1.wrong_rule_pack_routing_rate	wrong_routed_documents / documents_entering_routing_boundary	39.5%	
9	L1	Correct Abstention / Ambiguity Rate l1.correct_abstention_ambiguity_rate	correct_abstentions_or_flags / gold_ambiguity_cases	46.8%	
10	L1	Per-Type Precision & Recall l1.per_type_precision_recall	per document type (no single ratio)	—	computed per type
11	L2	Fact Recall l2.fact_recall	matched_expected_candidates / expected_extractable_candidates	42.9%	
12	L2	Fact Precision l2.fact_precision	matched_actual_candidates / scored_actual_candidates	55.1%	
13	L2	Value Accuracy l2.value_accuracy	value_correct_matched_candidates / matched_candidate_pairs	100.0%	
14	L2	Safe Rule Abstention Rate l2.safe_rule_abstention_rate	safe_abstentions / must_abstain_opportunities	99.7%	
15	L2	Candidate Evidence Traceability Rate l2.candidate_evidence_traceability_rate	candidates_with_valid_trace / candidates_requiring_provenance	88.7%	
16	L3	Field Mapping Accuracy l3.field_mapping_accuracy	correct_field_mappings / expected_mappable_facts	94.8%	
17	L3	Confirmed-Fact Preservation Rate l3.confirmed_fact_preservation_rate	preserved_confirmed_facts / confirmed_facts_in_scope	84.0%	
18	L3	Derived-Field Accuracy l3.derived_field_accuracy	correct_expected_derivations / expected_derivations	—	not computed:
19	L3	Transformation Completeness l3.transformation_completeness	required_targets_present / expected_transformable_targets	94.8%	
20	L3	Conflict & Provenance Safety l3.conflict_provenance_safety	passed_conflict_and_provenance_assertions / applicable_safety_assertions	100.0%	

#	LAYER	METRIC	NUMERATOR / DENOMINATOR	VALUE	
21	L4	Schema Pass Rate l4.schema_pass_rate	schema_valid_cases / cases_validated	82.5%	
22	L4	Required-Field Completeness l4.required_field_completeness	valid_required_fields_present / applicable_required_fields	100.0%	
23	L4	Invariant Pass Rate l4.invariant_pass_rate	passed_invariant_assertions / executed_invariant_assertions	—	not computed:
24	L4	Evidence-State Validity l4.evidence_state_validity	valid_evidence_state_assertions / applicable_evidence_state_assertions	100.0%	
25	L5	Verified Case Pass Rate l5.verified_case_pass_rate	strict_verified_case_passes / evaluated_cases	2.5%	
26	L5	Critical Field Accuracy l5.critical_field_accuracy	correct_final_tier1_fields / applicable_tier1_fields	63.9%	
27	L5	Critical Error Escape Rate ↓ l5.critical_error_escape_rate	machine_critical_errors_still_wrong_or_unresolved_at_exit / machine_critical_errors_present_at_review_entry	63.0%	
28	L5	Human Correction Burden ↓ l5.human_correction_burden	(corrected_machine_suggestions + manual_field_entries + manually_resolved_conflicts) / reviewed_fields	—	not computed:
D1	L5	Scripted Review Action Burden l5a.scripted_review_action_burden	L5-A diagnostic only; must not use the Human Correction Burden metric id	17.3%	

Naming note for engineers

The top-level README and the layer READMEs use newer names for five slots. This document uses the metrics-reference names, because they match the baseline report IDs.

#	METRICS REFERENCE / REPORT	NEWER README NAME
7	Wrong-Confident Classification Rate	Unsafe High-Score Classification Rate
8	Wrong Rule-Pack Routing Rate	Classification-Caused Wrong Routing Rate (+ Routing Registry Error Rate)
9	Correct Abstention / Ambiguity Rate	Safe Uncertainty Handling Rate
17	Confirmed-Fact Preservation Rate	Verified-Fact Preservation Rate
20	Conflict & Provenance Safety	Transformation Safety & Lineage

Sources: [Case_Intake_Evals_and_Testing_harness/Evals](#)
[Implementation/TAURI_INTAKE_EVAL_METRICS_REFERENCE.md](#),
[Evals_Illustrative_docs/metrics_golden-2026-09-25.md](#)

Scoring statuses

Every evaluated row has exactly one status. Only `SCORED` rows enter product metric denominators.

STATUS	MEANING	EFFECT ON PRODUCT METRICS
SCORED	Evaluated normally	Counts
NOT_APPLICABLE	The metric does not apply to this row; a zero denominator returns null	Shown as “—”, never 0 or 100%
NOT_SCORED_EXECUTION_ERROR	Harness or product failed to run (e.g. <code>TAURI_START_FAILED</code>)	Excluded; fails the infrastructure job
NOT_SCORED_CONTRACT_ERROR	Evaluation contract or compatibility mismatch (e.g. <code>SUT_COMPATIBILITY_MISMATCH</code>)	Excluded; routed to the contract owner
NOT_SCORED_FIXTURE_DEFECT	The test case itself is broken (e.g. <code>FIXTURE_ORACLE_HASH_MISMATCH</code>)	Excluded; fixture quarantined

Non-product defect taxonomy: `FIXTURE_DEFECT` , `EVALUATION_CONTRACT_ERROR` , `CHECKPOINT_CONTRACT_ERROR` , `HARNESS_EXECUTION_ERROR` .

Sources: `Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVALUATION_HARNESS_DESIGN.md` §3,
`Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVAL_METRICS_REFERENCE.md` ,
`Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVAL_REGRESSION_CI_DESIGN.md` §7, §12

Fixture gates, promotion and independence

What a test case must pass before it can score ARAM.

GATE	GATE	CHECKS
G0	Contract lock	Evaluation contract resolves and hash matches; SUT compatibility contract resolves; one owner per namespace; oracle-independence declaration passes.
G1	Fixture-spec integrity	Schema/constraints pass; Tier-1 authority valid; IDs unique and deterministic; fact/evidence/review states coherent.
G2	Source / build integrity	Required assertions injected exactly; no undeclared critical values; retained source matches rendered input.
G3	Rendered evidence integrity	Required evidence exists; artefact hash/page/span/slot resolve; rendered value matches the declared assertion.
G4	Mutation integrity	Mutation class/recipe declared; anchor transformations explicit; semantic/evidence effects match the declaration.
G5	Oracle integrity	Candidate matching resolves; oracle independently compiled; human-required resolution not guessed; enough truth for the target metric.
G6	Isolation / replay / package	Hidden answer key not visible to the SUT; reproducibility contract passes; package complete and hash-consistent.

Independence rules

- No fixture/oracle component may import, call, copy outputs from or reuse the production implementation of the behaviour it scores. Shared primitives are allowed only if they do not encode that behaviour.
- The oracle compiler consumes only the contract, the frozen `fixture_spec.yaml`, renderer anchors, mutation lineage and human-approved resolution. "It must not consume Tauri predictions."
- "No scored Tier-1 fact may be accepted solely because an LLM generated it." Authority must be `DETERMINISTIC_SYNTHETIC`, `HUMAN_APPROVED_SYNTHETIC` or `HUMAN_RESOLUTION_REQUIRED`.
- Tier-2/3 material may be sampled for review only if it is not scored.

Sources: `Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/EVALS_DATASET_AND_GROUND_TRUTH_DESIGN.md` §16, §18, §20–22;
`Case_Intake_Evals_and_Testing_harness/TAURI_INTAKE_DATASET_AGENT_DESIGN.md`
§7.3, §34, §37, §39, §41–42

Run manifest, replay and baselines

What is recorded so that any result can be reproduced.

Run manifest fields

run_id fixture_id fixture_version fixture_spec_hash
evaluation_contract_version evaluation_contract_hash
sut_compatibility_version intake_version eval_mode_version
harness_version mode layer platform injected_oracle_hashes

Scored results additionally pin:

checkpoint_versions oracle_compiler_version oracle_hashes
candidate_matching_policy_version metric_catalog_version
comparator_policy_version scoring_status

Replay modes

- `BYTE_REPLAYABLE` — all replayed artefact hashes equal the manifest hashes
- `SEMANTICALLY_REPLAYABLE` — byte identity not guaranteed, but spec hash, assertion/candidate set, semantic provenance and oracle values must match
- `FROZEN_ONLY` — approved artefacts are never regenerated; stored hashes must match the frozen manifest

A replay creates a new run id and links `replay_of` .

Run modes

- `isolated` — one layer with known-good injected input
- `end_to_end` — complete workflow, no injection
- `scripted_review` — L5-A: deterministic reviewer script
- `interactive_review` — L5-B: real reviewer telemetry under a declared protocol

Baseline comparison fields

new_failures fixed_failures persistent_failures fixture_defects contract_errors metric_deltas

“A baseline is a scored result over a frozen dataset and pinned evaluation semantics.” It pins the regression dataset version, fixture/oracle hashes, evaluation-contract version/hash, candidate-matching version, metric-catalogue version, comparator version and intake version.

Sources: `Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_TEST_HARNESS_DESIGN.md` §3, §5, §8, §10;
`Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVALS_L0_L5_IMPLEMENTATION_DESIGN.md` §14;
`Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVAL_REGRESSION_CI_DESIGN.md` §3, §9

Failure and root-cause taxonomy

How failures are named, and how their origin is recorded.

Product failure families

- RECOVERY_FAILURE
- CLASSIFICATION_FAILURE
- WRONG_CONFIDENT_CLASSIFICATION
- WRONG_RULE_PACK
- EXTRACTION_MISS
- EXTRACTION_SPURIOUS
- VALUE_ERROR
- PROVENANCE_ERROR
- MAPPING_ERROR
- DERIVATION_ERROR
- CONFLICT_HANDLING_ERROR
- SCHEMA_ERROR
- INVARIANT_ERROR
- HUMAN_REVIEW_ESCAPE

End-to-end (L5) failure types

- FINAL_CASE_VALUE_ERROR
- FINAL_CASE_STATE_ERROR
- CRITICAL_CONFLICT_LOST
- HUMAN_REVIEW_ESCAPE
- REVIEW_TRIGGER_ERROR
- FINAL_EXPORT_ERROR

Non-product defects

- FIXTURE_DEFECT
- EVALUATION_CONTRACT_ERROR
- CHECKPOINT_CONTRACT_ERROR
- HARNESS_EXECUTION_ERROR

“Do not assign product root-cause layers to unscorable fixtures.”

L5 containment record

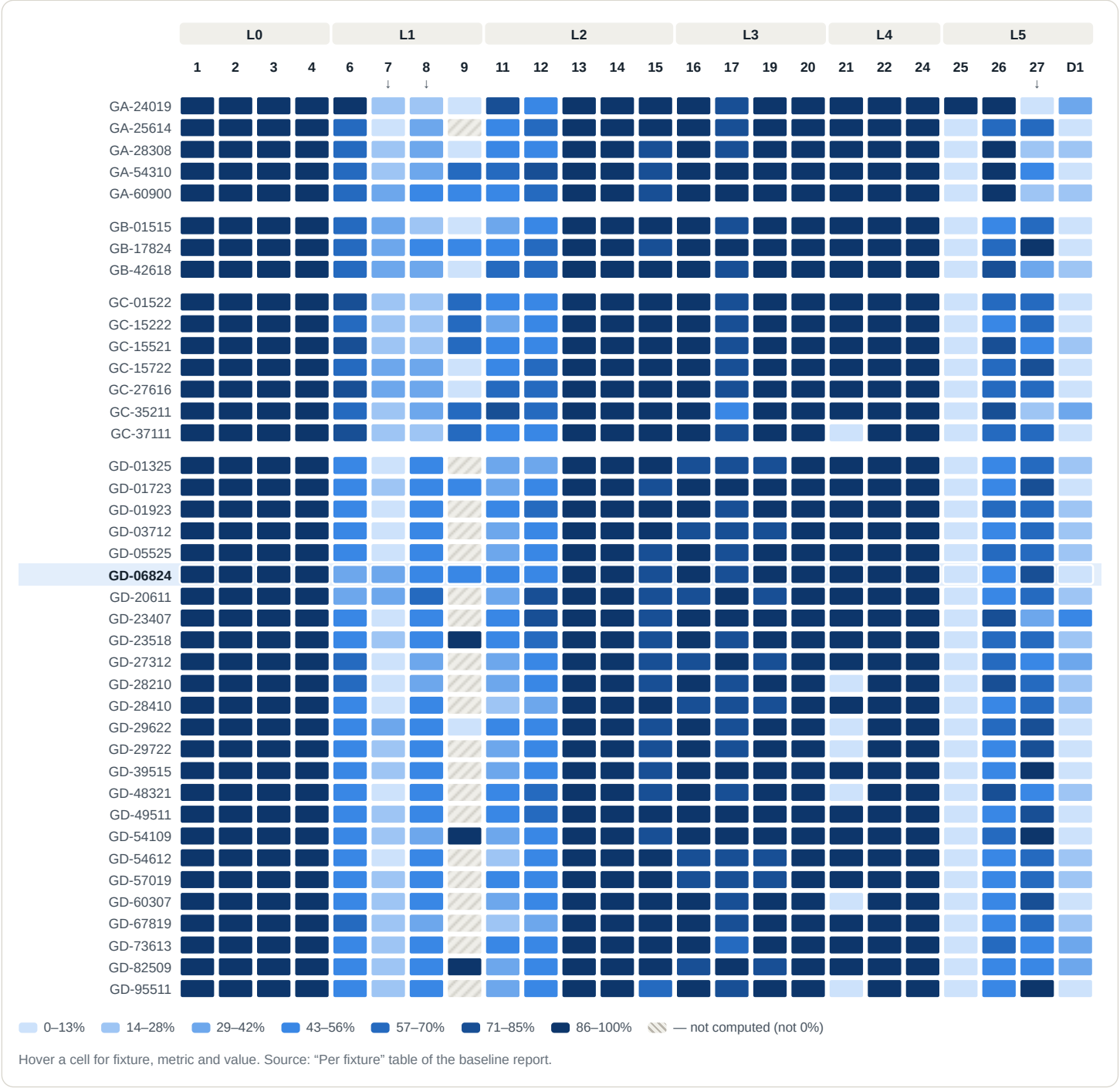
- origin_layer
- origin_component
- first_detection_opportunity
- detection_layer
- detection_component
- expected_containment_layer
- expected_containment_action
- actual_containment_layer
- actual_containment_action
- escape_boundary
- final_impact

Root cause = the **earliest causally sufficient divergence**, found by walking explicit lineage. “Do not infer causality from layer ordering alone.” Contributing causes are stored separately. At L5, a single `root_cause_layer` is not enough: origin, detection opportunity, expected and actual containment, and escape boundary are all recorded.

Sources: `Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVALS_L0_L5_IMPLEMENTATION_DESIGN.md` §9, §11;
`Case_Intake_Evals_and_Testing_harness/Evals`
`Implementation/TAURI_INTAKE_EVALUATION_HARNESS_DESIGN.md` §12–14

Per-fixture results (40 fixtures)

Each cell is that fixture's own pooled percentage for one metric (column numbers match Appendix A). Darker = higher percentage. For ↓ columns, higher is worse. Hatched = not computed. The highlighted row is CASE-GD-06824 .



Show as table

FIXTURE	1	2	3	4	6	7↓	8↓	9	11	12	13	14	15	16	17	19	20	21	22	24	25	26	27↓	D1
GA-24019	100.0	100.0	100.0	100.0	100.0	20.0	20.0	0.0	83.3	55.6	100.0	87.5	100.0	100.0	83.3	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
GA-25614	100.0	100.0	100.0	100.0	71.4	0.0	28.6	—	45.5	71.4	100.0	100.0	100.0	100.0	81.8	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	66.7
GA-28308	100.0	100.0	100.0	100.0	71.4	16.7	37.5	0.0	44.4	44.4	100.0	92.9	80.0	100.0	77.8	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	88.9
GA-54310	100.0	100.0	100.0	100.0	60.0	25.0	37.5	66.7	66.7	75.0	100.0	100.0	85.7	100.0	88.9	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	87.5

Fixture	1	2	3	4	6	7	8	9	11	12	13	14	15	16	17	19	20	21	22	24	25	26
GA-60900	100.0	100.0	100.0	100.0	60.0	40.0	44.4	50.0	55.6	71.4	100.0	100.0	83.3	100.0	88.9	100.0	100.0	100.0	100.0	100.0	0.0	88.9
GB-01515	100.0	100.0	100.0	100.0	70.0	30.0	27.3	0.0	33.3	57.1	100.0	100.0	100.0	100.0	83.3	100.0	100.0	100.0	100.0	100.0	0.0	55.6
GB-17824	100.0	100.0	100.0	100.0	60.0	40.0	42.9	50.0	50.0	71.4	100.0	100.0	83.3	100.0	90.0	100.0	100.0	100.0	100.0	100.0	0.0	62.9
GB-42618	100.0	100.0	100.0	100.0	70.0	41.7	30.8	0.0	58.3	63.6	100.0	100.0	87.5	100.0	83.3	100.0	100.0	100.0	100.0	100.0	0.0	80.0
GC-01522	100.0	100.0	100.0	100.0	75.0	18.2	20.0	66.7	46.2	54.5	100.0	100.0	100.0	100.0	84.6	100.0	100.0	100.0	100.0	100.0	0.0	60.0
GC-15222	100.0	100.0	100.0	100.0	66.7	20.0	26.7	66.7	30.8	57.1	100.0	100.0	100.0	100.0	84.6	100.0	100.0	100.0	100.0	100.0	0.0	50.0
GC-15521	100.0	100.0	100.0	100.0	75.0	18.2	20.0	66.7	53.8	53.8	100.0	100.0	87.5	100.0	84.6	100.0	100.0	100.0	100.0	100.0	0.0	72.2
GC-15722	100.0	100.0	100.0	100.0	66.7	33.3	40.0	0.0	46.2	66.7	100.0	100.0	100.0	100.0	84.6	100.0	100.0	100.0	100.0	100.0	0.0	60.0
GC-27616	100.0	100.0	100.0	100.0	72.7	33.3	35.7	0.0	58.3	63.6	100.0	100.0	100.0	100.0	83.3	100.0	100.0	100.0	100.0	100.0	0.0	70.0
GC-35211	100.0	100.0	100.0	100.0	71.4	16.7	30.0	66.7	75.0	60.0	100.0	100.0	100.0	100.0	50.0	100.0	100.0	100.0	100.0	100.0	0.0	85.0
GC-37111	100.0	100.0	100.0	100.0	75.0	18.2	20.0	66.7	53.8	53.8	100.0	100.0	87.5	100.0	84.6	100.0	100.0	0.0	100.0	100.0	0.0	70.0
GD-01325	100.0	100.0	100.0	100.0	55.6	0.0	44.4	—	30.0	33.3	100.0	100.0	100.0	85.7	80.0	85.7	100.0	100.0	100.0	100.0	0.0	55.6
GD-01723	100.0	100.0	100.0	100.0	54.5	25.0	46.2	50.0	40.0	46.2	100.0	100.0	75.0	87.5	86.7	87.5	100.0	100.0	100.0	100.0	0.0	50.0
GD-01923	100.0	100.0	100.0	100.0	57.1	0.0	42.9	—	50.0	66.7	100.0	100.0	100.0	87.5	83.3	87.5	100.0	100.0	100.0	100.0	0.0	70.0
GD-03712	100.0	100.0	100.0	100.0	50.0	0.0	50.0	—	36.4	57.1	100.0	100.0	100.0	85.7	81.8	85.7	100.0	100.0	100.0	100.0	0.0	55.6
GD-05525	100.0	100.0	100.0	100.0	57.1	0.0	42.9	—	36.4	50.0	100.0	100.0	80.0	87.5	81.8	87.5	100.0	100.0	100.0	100.0	0.0	70.0
GD-06824	100.0	100.0	100.0	100.0	41.7	37.5	50.0	50.0	42.9	54.5	100.0	100.0	75.0	87.5	85.7	87.5	100.0	100.0	100.0	100.0	0.0	50.0
GD-20611	100.0	100.0	100.0	100.0	33.3	33.3	66.7	—	33.3	75.0	100.0	100.0	75.0	85.7	88.9	85.7	100.0	100.0	100.0	100.0	0.0	44.4
GD-23407	100.0	100.0	100.0	100.0	50.0	0.0	50.0	—	50.0	75.0	100.0	100.0	75.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	0.0	75.0
GD-23518	100.0	100.0	100.0	100.0	44.4	20.0	45.5	100.0	50.0	60.0	100.0	100.0	85.7	87.5	83.3	87.5	100.0	100.0	100.0	100.0	0.0	60.0
GD-27312	100.0	100.0	100.0	100.0	60.0	0.0	40.0	—	33.3	42.9	100.0	100.0	75.0	85.7	88.9	85.7	100.0	100.0	100.0	100.0	0.0	66.7
GD-28210	100.0	100.0	100.0	100.0	71.4	0.0	28.6	—	41.7	55.6	100.0	100.0	83.3	100.0	83.3	100.0	100.0	0.0	100.0	100.0	0.0	77.8
GD-28410	100.0	100.0	100.0	100.0	50.0	0.0	50.0	—	20.0	33.3	100.0	100.0	100.0	85.7	80.0	85.7	100.0	100.0	100.0	100.0	0.0	55.6
GD-29622	100.0	100.0	100.0	100.0	46.2	40.0	53.3	0.0	42.9	54.5	100.0	100.0	85.7	100.0	85.7	100.0	100.0	0.0	100.0	100.0	0.0	60.0
GD-29722	100.0	100.0	100.0	100.0	50.0	20.0	50.0	—	30.0	42.9	100.0	100.0	75.0	100.0	80.0	100.0	100.0	0.0	100.0	100.0	0.0	50.0
GD-39515	100.0	100.0	100.0	100.0	50.0	16.7	50.0	—	33.3	55.6	100.0	100.0	83.3	87.5	86.7	87.5	100.0	100.0	100.0	100.0	0.0	50.0
GD-48321	100.0	100.0	100.0	100.0	57.1	0.0	42.9	—	54.5	60.0	100.0	100.0	85.7	100.0	81.8	100.0	100.0	0.0	100.0	100.0	0.0	77.8
GD-49511	100.0	100.0	100.0	100.0	50.0	20.0	50.0	—	45.5	62.5	100.0	100.0	100.0	100.0	90.9	100.0	100.0	100.0	100.0	100.0	0.0	44.4
GD-54109	100.0	100.0	100.0	100.0	54.5	14.3	41.7	100.0	40.0	46.2	100.0	100.0	85.7	87.5	86.7	87.5	100.0	100.0	100.0	100.0	0.0	60.0
GD-54612	100.0	100.0	100.0	100.0	57.1	0.0	42.9	—	27.3	50.0	100.0	100.0	100.0	85.7	81.8	85.7	100.0	100.0	100.0	100.0	0.0	55.6
GD-57019	100.0	100.0	100.0	100.0	42.9	25.0	57.1	—	44.4	50.0	100.0	100.0	100.0	83.3	77.8	83.3	100.0	100.0	100.0	100.0	0.0	50.0
GD-60307	100.0	100.0	100.0	100.0	55.6	16.7	44.4	—	38.5	55.6	100.0	100.0	100.0	100.0	84.6	100.0	100.0	0.0	100.0	100.0	0.0	55.6
GD-67819	100.0	100.0	100.0	100.0	62.5	16.7	37.5	—	20.0	33.3	100.0	100.0	100.0	100.0	80.0	100.0	100.0	100.0	100.0	100.0	0.0	50.0
GD-73613	100.0	100.0	100.0	100.0	50.0	25.0	50.0	—	42.9	50.0	100.0	100.0	100.0	100.0	71.4	100.0	100.0	100.0	100.0	100.0	0.0	66.7
GD-82509	100.0	100.0	100.0	100.0	42.9	25.0	44.4	100.0	37.5	50.0	100.0	100.0	100.0	85.7	87.5	85.7	100.0	100.0	100.0	100.0	0.0	55.6
GD-95511	100.0	100.0	100.0	100.0	50.0	16.7	50.0	—	35.7	45.5	100.0	100.0	71.4	100.0	85.7	100.0	100.0	0.0	100.0	100.0	0.0	55.6

Glossary: engineering term → plain wording

Customer wording first; the engineering term is what appears in logs, reports and the repository.

ENGINEERING TERM	PLAIN WORDING
fixture	controlled test case
fixture_spec.yaml	authoritative test-case definition
oracle	independently derived expected result (answer key)
SUT	ARAM Intake, the system under test
sut_visible	the only files ARAM is allowed to see
Test Harness	the system that runs the controlled test and records it
Evaluation Harness	the system that independently compares and scores the result
checkpoint	recorded result at a specific processing stage
provenance	trace back to the supporting evidence
abstention	correctly declining to make an unsupported determination
invariant	a condition that must always hold for a valid case
Tier-1	a fact whose error can change route, gate, deadline or case correctness
H1	human verification of each proposed value
L5-A	scripted end-to-end regression review
L5-B	real-human reviewer study
NOT_SCORED_FIXTURE_DEFECT	test-case defect: excluded from product scoring
NO_ELIGIBLE_UNITS	nothing to measure in this run: shown as “—”
baseline	a scored result over a frozen dataset with pinned evaluation rules

ARAM Intake · Evaluation & Testing Harness · Report numbers: baseline `golden-2026-09-25` (Evals_Illustrative_docs/metrics_golden-2026-09-25.md). All case material is synthetic: no real person, no real matter. This document describes tested system behaviour. It does not assess legal correctness and is not legal advice.